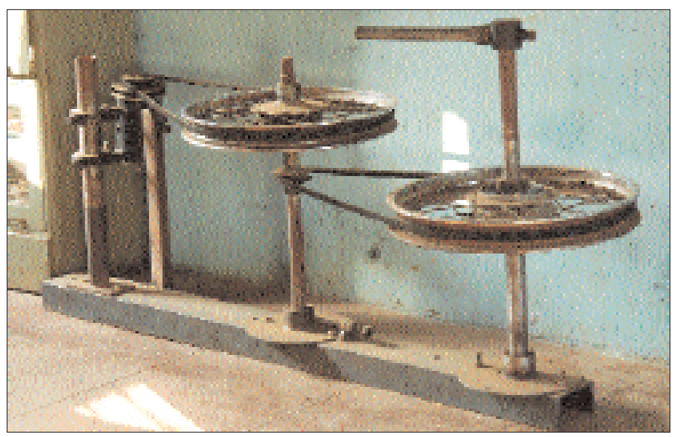
It's interesting that the students are given a free hand to explore and discover their XO. When we were at the school, we saw many students taking photos and video clips of us!

We saw quite a few animated clips that Mr Surve had prepared—on various topics such as the earth's orbit and the water cycle system—which were distributed to those in the class on their XOs. The laptops act as individual nodes and create a wireless network, also known as a *mesh*, when two or more XOs are in a 1 km radius. This can be used to chat and transfer files without the need for an Internet connection. We found Surve using the chat application innovatively by asking the class a question that required a written reply. The students would reply via the chat tool. The kids are taught how to write Marathi words in eToys, an application similar to MS Paint, and they complete their homework on Write, an MS Word-like application. According to Surve, the students are now eager to come to school because they get to learn something new about their laptop every day, either from him or from their classmates. Parents, who were earlier not interested in sending their kids to school, have changed their attitude because they know that their kids are getter a good level of education, thanks to the XO.

There are, however, hurdles to overcome. Foremost amongst these is the availability of electricity, which, of course, is needed for charging the laptops. There are plans to get solar panels for recharging the batteries. There also happens to be a contraption to aid in recharging



The Cow Power charging device, stowed away for later use

the laptops, which has been developed using local materials and runs on cow power! A cow moves in a circle, driving a system of belts and pulleys connected to a dynamo to generate power to charge all the laptops. The dynamo is the one used in Fiat cars, very commonly seen as taxis in Mumbai. It takes one to two hours to charge all the 20 laptops in the village simultaneously. This is possible due to the extremely low power consumption of the laptop and the robust power supply built into it. The contraption is the brainchild of Arjun Sarwal, an OLPC volunteer.

The XO has a robust frame, but we noticed a few laptops without antennas, and two non-functioning ones. The OLPC volunteers who accompanied us said they would be replaced. We also saw that the keyboards were in English; this might, we thought, initially hinder students from working on the XO; plans are under way to create a keyboard with keys in the Devanagri script, but it's not clear when it will be introduced.

There are plans to get solar panels for recharging the batteries. For now, the laptops are recharged using a contraption that turns cow power to electricity

The laptops the students have been given are early prototypes of the XOs available today, and are not likely to be replaced, though.

Q&A

We were looking to get some questions answered by the OLPC India team. For example, we wanted to know what the roadmap for the programme was, how they were planning on taking the programme to other parts of the country, and so on. The questions below were answered by

Manusheel Gupta (MG), Technical Consultant, One Laptop Per Child Inc; Sumit Chowdhury (SC), CIO, Reliance Communications; Carla Gomez Monroy (CGM), OLPC India team member; and Amit Gogna (AG), OLPC India team member.

What is being taught in Khairat School using the OLPC laptops? Is there any courseware available for it?

AG: We're working on a booklet for teachers with a basic introduction to activities on the XO. It's too early to talk about courseware. The XO gives teachers the liberty to create projects and use various "activities" that suit their needs as they go about with their daily curriculum.

How long do the laptops have to be cranked before they are fully charged?

MG: We have several research groups looking at different human-power solutions, including a hand crank, a foot treadle, a yo-yo, and a pulley system. Our goal is a minimum of a 1:10 ratio of "cranking"—one minute of cranking should give you 10 minutes of use. The XOs can also be charged using a solar device, which is a good solution for India.

AG: The laptops currently shipped to us do not come with the crank mechanism. They get charged using an AC adaptor. In addition, one student at the school uses solar panels to charge his XO.

SC: We do not have laptops that have to be cranked. Getting more solar-powered battery chargers seems to be a more effective alternative.

The XO is aimed at students of what standard (or class, or grade)?

CGM: It is targeted at primary school children. In most countries, children start primary school at the age of five. However, in some schools, there are children who could not start school at that young an age, which means that in some cases we have children as old as 15 or 16 in primary school.

How do you plan to introduce the OLPC concept in classrooms across India, where the average head-count per class is 60?

SG: We will look to adopt entire schools or communities with the help of the Digital Bridge Foundation—comprising corporate partners, non-profit organisations, ministers at State level, and most importantly, interested volunteers—and try to provide laptops to all the students.

AG: At present, we intend to start with rural schools. We are in talks with various state governments, NGOs, and corporate bodies to assist us. We are building an ecosystem for the collection of funds, and are working with educationists and interested volunteers to create educational content and plan for its deployment into the system.

CGM: The issue is not only the size of the class, but also how the laptops are used to foster the children's learning.

Don't you think that donating only to governments is impeding the progress of the project?

MG: Our goal is to reach the rural children of developing countries. Dealing with governments makes this process efficient. However, this is not the only strategy that we are adopting. We have various running programmes—like "Give 1 Get 1" and "Give Many," where the purchase made by an individual or organisation benefits rural parts of the world. You can find more information at http://tinyurl.com/yo4s3h.

CGM: The idea was to have governments sponsoring for their people. In most countries, governments run the entire educational system, especially for public schools. It is easier to work with one well-established and influential entity in a country than to create ecosystems in each country to make it happen. That's why the first approach was through the governments.

"The XO laptops are aimed at children in specific age ranges, and contain activities tailored for them. It would be unwise to compare it to the ASUS Eee"

The students just can't get enough of the XO

The landing cost of the XO in India is about Rs 8,500. Taking that into consideration, don't you think the ASUS Eee PC is a more viable proposition? Though it does not support the ability to create mesh networks on the fly, it supports an interface that teachers are familiar with (MS Windows)...

SC: We believe in following the constructionist approach to teaching. The XO laptops are aimed at children in specific age ranges, and contain activities tailored for them. It would be unwise to compare the two.

AG: Comparing XOs to other laptops defeats the whole purpose. Remember, we are targeting young minds and want to create a system that triggers their thinking and creative abilities and not one that turns them into experts at Word, Excel, or PowerPoint. Some key features that stand out are that the laptops are very sturdy, consume low power, and foster collaborative learning through the mesh network.

What happens if a student says a laptop is lost? Do you provide for a replacement?~

MG: That depends on the definition of "lost." If a laptop is stolen, the security mechanism enabled on the laptop will disable it within hours.

AG: The kids are given some basic instructions about handling the laptops, thus minimising the chances of it getting lost. Having said that, some laptops are needed as spares for compensating for rough handling by the kids.

Earlier in the year, there were reports of students in some countries being caught viewing pornography on their XOs. What has been done since to combat that problem?

SC: The porn menace can be easily solved in India. Most rural villages do not have access to the Internet. When we do provide them with Internet access, we can have filters installed to block pornographic and malicious Web sites.

In one of his mailing list posts, Gogna said: "OUR OLPC MESH NETWORK is going to set its NODES all across the NATION." Does this mean that Reliance ADAG will provide the backbone network connectivity for the OLPC project in India? If yes, by when will this be completed?

MG: We're glad to have seen the best support from Reliance ADAG in the first OLPC pilot project conducted in Khairat School. We don't know the answer, and will take some time depending upon various parameters. We'd be happy if Reliance ADAG provides us with the networking services.

AG: That statement was much more of an analogy than fact. What I meant by "OLPC Mesh Network" was the formation of the ecosystem for the project to spread across various parts of the country, acting as nodes.

CGM: The analogy comes from how the actual Mesh Network of the XO laptops works. The XO laptops can connect to each other (without being connected to the Internet) and thus collaborate on all the different software applications ("activities"). The XOs can also be nodes that replicate the Internet signal among them, in case only one of them has access to the Internet. They can do it even when turned off.

What are the goals (and roadmap) of the OLPC project in India?~

MG: The OLPC project in India and all around the world aims to revolutionise the way we teach children. We hope to reach the remotest parts of India by means of the "Digital Bridge Foundation," comprising corporate partners, non-profit organisations, Ministers at the state level, and most importantly, interested volunteers.

AG: We also intend to reach most villages and towns across the country.

The OLPC project seems to have started off on the right foot in India. There are loose ends, however; foremost amongst these is the problem of creating awareness about the project. The OLPC team is indeed working with state governments to provide support to schools in Gujarat and Kolkata. We'll be keeping our eyes open. ■

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